Evaluating causal discovery methods on real measurements: cause–effect pairs, protein signalling and physical devices

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Abstract

Causal discovery methods are usually validated on simulated data. We evaluate twenty-two methods on real data with a known or consensus structure: the Tübingen cause–effect pairs, the protein-signalling data of Sachs et al., and recordings from the Causal Chambers, physical devices with a known graph. Twelve methods are established (PC, GES, DirectLiNGAM, NOTEARS, DAGMA and PCMCI+ for graphs; IGCI, RECI, ANM-HSIC, a pairwise LiNGAM rule, a Gaussian-process ANM and LOCI for pairs), four are re-implementations of 2025–2026 preprints (compression, topological residual asymmetry, BF-BIC with BOSS, SNOE-style orientation) and six were developed for this study and frozen before evaluation: sequential certificates for edges and directions, PC with a nonparametric test, a block-replicated additive method (BTAD) and two classifiers trained on simulated data. On the Causal Chambers pairs RECI has the highest accuracy (**97.9%** and **94.3%** on two sets), not distinguishable from IGCI; on Tübingen no method is distinguishable from RECI (**60.8%**) except the classifier PDM, which is below chance (**40.2%**, $\boldsymbol{p = 0.059}$) although it reaches **95%** on simulated pairs; all methods are within the sampling error of chance on **17** Sachs pairs. For graphs, the strict skeleton $\boldsymbol{F_1}$ of DirectLiNGAM and GES is highest on the chamber recordings with independent rows (**0.55** on two recordings, **0.70** on two others), the methods we developed are ahead on the Sachs data ($\boldsymbol{F_1}$ **0.62** to **0.67** against **0.58** for PC and GES, a difference of one to two of **17** reference edges) and on the walk recordings (**0.42** against **0.38**), and no method exceeds a strict $\boldsymbol{F_1}$ of **0.42** on the seventeen random-walk recordings. Fifteen wind-tunnel recordings are random walks with an effective sample size of about three observations out of **4000**, and a test that assumes independent rows rejects **4.8** times as many graph-implied independences as a block test there. The sequential certificates keep false edges rare in simulation under repeated inspection and nonlinear mechanisms, but not on the chambers. We state which choices

were fixed before and after seeing the data, and the limits of the evidence: few reference edges, few recordings per group, and several methods not evaluated.

Keywords: causal discovery, benchmark, real data, cause–effect pairs, Causal Chambers, serial dependence

MSC Classification: 62H22 , 62G10 , 68T05

1 Introduction

Methods for learning causal structure from observational data are mostly validated on simulated data, where the generating mechanism is known and the assumptions of the method can be matched to it. Real measurements differ in ways that simulation rarely reproduces: mechanisms are nonlinear and heterogeneous, sensors are collinear, unobserved processes induce dependence between measured variables, and rows are often not independent draws. Benchmarks with real data and a known structure exist, but each covers a narrow slice: the Tübingen cause–effect pairs (Mooij, Peters, Janzing, Zscheischler, & Schölkopf, 2016), the protein-signalling data of Sachs, Perez, Pe'er, Lauffenburger, and Nolan (2005), whose reference network is a contested consensus (Mooij, Magliacane, & Claassen, 2020), and, more recently, the Causal Chambers, physical devices whose graph is known from their construction (Gamella, Peters, & Bühlmann, 2025).

This paper asks a plain question: when methods are applied unchanged to several real data families, which ones work, and do recent ideas beat simple baselines? We evaluate twenty-two methods on four families of real data, under a protocol in which every method and every setting was fixed and recorded before the data it is scored on were run, with the exceptions listed in Section 2.5. The methods are twelve established methods (PC, GES, DirectLiNGAM, NOTEARS, DAGMA and PCMCI+ for graphs; IGCI, RECI, ANM-HSIC, a pairwise LiNGAM rule, a Gaussian-process ANM and LOCI for pairs), four recent ones that we re-implemented from 2025–2026 preprints (a sorted-conditional compression rule, topological residual asymmetry, BF-BIC with BOSS, and an edge-orientation rule in the style of SNOE) and six methods that we developed during the project that produced this study: sequential certificates for adjacency and direction, PC with a nonparametric test, a block-replicated additive method, and two classifiers trained on simulated data. Including our own methods is deliberate. Several were designed to address failures of the baselines, and the comparison tells whether the design worked on measurements.

Contributions.

1. A multi-dataset evaluation with a recorded protocol. It covers 102 Tübingen pairs, 216 pair analyses of edges of the Causal Chambers, 17 Sachs pairs, the Sachs data, and 21 chamber recordings with a known graph, and it reports a strict score against the reference graph and a score adjusted for dependences the graph cannot explain.

2. The result that, on pairs, a regression-error rule (RECI) has the highest accuracy on the Causal Chambers pairs (not distinguishable from IGCI), that on Tübingen no method is distinguishable from RECI except a classifier trained on simulated pairs, which is accurate on its simulated test set and below chance on Tübingen (Section 4.1).
3. The result that, on graphs, the ranking of methods depends on the data family: DirectLiNGAM and GES on chamber recordings with independent rows, some of our methods by a margin of one or two edges on the Sachs data, and no method above a strict F_1 of 0.42 on the random-walk recordings (Section 4.2).
4. A diagnosis of why the walk recordings defeat every method: their exogenous variables are random walks with an effective sample size of a few observations, so a test that assumes independent rows rejects 4.8 times as many graph-implied independences as a block test on the wind-tunnel walks, and a reference set built with it is contaminated (Section 4.3).
5. The places where our own methods do help: control of false edges under repeated inspection and under nonlinear mechanisms in simulation (Section 4.4).

What the paper does not claim.

It does not claim that the methods we did not evaluate (QCCD, CAM and amortised methods such as AVICI (Lorch, Sussex, Rothfuss, Krause, & Schölkopf, 2022), whose package we could not run with a compatible version of JAX) fail on these data, and it does not claim that the data sets are representative of causal discovery in general. Several data sets have few reference edges, so most differences between methods are within sampling error, and we give intervals where we can.

2 Data and protocol

2.1 Data

Tübingen pairs.

The 102 scalar-scalar pairs of the cause–effect benchmark (Mooij et al., 2016), with the official weights. Tied values are dequantised by adding uniform noise of the width of the smallest gap, long pairs are subsampled to 2000 rows, and the preprocessing seed is fixed.

Causal Chambers.

The wind tunnel and the light tunnel (Gamella et al., 2025) are physical devices with a graph given by their construction. We use recorded experiments in which the actuators vary, and the nodes of the chamber graph with at least 20 distinct values. Three kinds of recordings occur. In the *i.i.d.* recordings the actuators are redrawn independently at random: the wind tunnel with random intake loads and the light tunnel with uniform inputs, used for the pair and graph evaluations (two recordings with ten subsamples each). In the *walk* recordings the actuators follow random walks: 15 runs of the wind tunnel, the pressure-control experiment, and three light-tunnel experiments, of which one (`color_mix`) is a walk and two have nearly independent

rows (Section 4.3). The graphs have 8 to 39 edges among 10 to 20 variables and are bipartite: settings point to measurements. The pair sets are the directed edges of these graphs: 47 distinct edges of the two i.i.d. recordings, each analysed on two subsamples (94 pair-runs, of which about 47 are independent), and 122 edges of five further recordings (the first recording of each configuration: 26 from one wind-tunnel walk, 18 from the pressure-control experiment, 37 and 27 from the two light-tunnel recordings with nearly independent rows, 14 from the light-tunnel walk). Because causes are actuators with simple marginals, these pair sets are structurally easy.

Sachs.

The 853 observational cells with 11 phosphoproteins, log-transformed; the reference network is the 17-arc consensus of Sachs et al. (2005), which is contested (Mooij et al., 2020). For the graph benchmark we use all cells; for pairs we use its 17 directed arcs as a pair set.

2.2 Row order

Rows are shuffled at random in the analyses of the recordings with independent actuators, of the Sachs cells, and of all pair sets, so that serial dependence of the sensors (dynamics, hysteresis) is broken by construction; the label "i.i.d." for these recordings means that the actuators are redrawn independently, not that independence of rows was verified. The graph benchmarks on the nineteen new recordings (Section 4.2) use the first 4000 rows in time order, and the earlier false-edge counts for the sequential certificate on the new recordings used shuffled rows.

2.3 Scores

Pairs. Forced-choice accuracy of the direction (unweighted, and weighted by the Tübingen weights), with Clopper–Pearson intervals; and accuracy at a given coverage when pairs are ranked by a method's confidence.

Graphs. Precision, recall and F_1 of the skeleton against the reference graph ("strict"). Because the chamber graphs do not explain all dependence among the sensors (for instance four barometers share an ambient pressure), we also report precision against the graph plus a reference set V of non-adjacent pairs whose graph-implied conditional independence is rejected on held-out rows ("adjusted"); the strict score is the headline.

2.4 Effective sample size

For each recording we measure the lag-1 autocorrelation ρ of the exogenous variables and the implied effective sample size $n(1-\rho)/(1+\rho)$ of the first 4000 rows. This is the stationary AR(1) formula, a heuristic for random walks, and it is not a test; the effective sample size also coincides with the device (fifteen of the seventeen low values are wind-tunnel recordings and the two high values are light-tunnel recordings). The result (Section 4.3) splits the nineteen new recordings into two with an effective sample size above 1000 and seventeen below 20.

2.5 Protocol and provenance

Methods, settings and the scripts that score them were committed to the repository before the run in every case except the following, which are post hoc and flagged where they occur. First, the Causal Chambers recordings with independent rows and the Sachs data were inspected while we were developing the sequential certificates, so those certificates were adjusted after seeing them (a nonparametric test, a larger conditioning order). Second, the nine simulated pair families used to choose the agreement rule of the direction certificate are design data. Third, the reference set V was introduced after seeing false edges. Every other number is from a frozen run: the new chamber recordings, the Tübingen pairs for all methods, the learned models (trained only on simulated data and frozen before evaluation), the baselines, and the Sachs run of all methods. The six methods of the study and their settings are described in Section 3 and the appendix.

3 Methods compared

3.1 Established baselines

Graphs. PC with Fisher-z tests at $\alpha = 0.05$ and 0.01 (Spirtes, Glymour, & Scheines, 2000), GES with the BIC score (Chickering, 2002), DirectLiNGAM (Shimizu et al., 2011) with edges thresholded at $|B| > 0.05$, all from `causal-learn`; NOTEARS with a linear model (Zheng, Aragam, Ravikumar, & Xing, 2018) (our compact re-implementation, ℓ_1 penalty 0.1, threshold 0.3); DAGMA, linear with the ℓ_2 loss (package `dagma`, $\lambda_1 = 0.03$, threshold 0.3); and PCMCI+ (Runge, 2020) from `tigramite` with partial-correlation tests, lags 0 and 1, $\alpha = 0.01$, taking the contemporaneous links as the skeleton. We score skeletons only.

Pairs. IGCI in its entropy form with a uniform reference (Daniušis et al., 2010), RECI, which prefers the direction with the smaller regression error after min–max scaling (Blöbaum, Janzing, Washio, Shimizu, & Schölkopf, 2018), an additive noise model with a spline regression and an HSIC independence criterion (ANM-HSIC; Hoyer, Janzing, Mooij, Peters, and Schölkopf 2009), and the first-order pairwise LiNGAM likelihood-ratio statistic (Hyvärinen & Smith, 2013). These are short re-implementations of the published rules and were not tuned. We add a Gaussian-process additive noise model (`causal-learn`, direction with the larger independence p-value, 400 rows) and LOCI (Immer et al., 2023), run with the authors' public code in its default setting (neural network, 5000 steps, HSIC independence test, at most 1000 rows per pair).

Recent methods (2025–2026), re-implemented.

Four preprints were implemented from their descriptions, with settings fixed before the evaluation (experiments 42–44); none has public code. *Sorted-conditional compression* (Stienstra, 2026): both variables are quantised to 256 levels, the rows are sorted by the candidate cause, the other variable is differenced and compressed with bzip2, and the direction with the smaller normalised length wins; we run it on the rows as given and on our dequantised rows. *Topological residual asymmetry* (TRA; El Bouchattaoui

2026): cross-fitted smoothing-spline residuals, a rank-uniform point cloud, and the total zero-dimensional persistence in a window; the constants ($\kappa = 1$, $c_\beta = 4$, $K = 5$ folds, at most 1000 rows) are our choices because the preprint does not fix them, and 20 Tübingen pairs gave no value. *BF-BIC with BOSS* (Ramsey, Andrews, & Spirtes, 2025): each variable is expanded in Legendre polynomials of degree 3, the local BIC is summed over the child's basis columns (standard Gaussian likelihood, since the preprint's formula has an inconsistency), and a permutation search of the best-order type (one random start) returns the skeleton. *SNOE-style orientation* (Huang & Zhou, 2025): every true edge of the reference graph is oriented by the larger additive-noise log-likelihood of (child | cause, other neighbours) plus (cause | its neighbours), with spline regression; the neighbours come from the reference graph, which favours the rule, and the sequential ranking of the preprint is not implemented.

3.2 Methods developed for this study

All six were frozen before their evaluation on the data in which they are scored, except where Section 2.5 says otherwise. Details are in the appendix.

Direction certificate (relative fit).

For a pair, two forecasters of the joint law, one per factorisation, are fitted sequentially on past rows; the running mean of the log-score difference has an asymptotic confidence sequence (Waudby-Smith, Arbour, Sinha, Kennedy, & Ramdas, 2024); a direction is reported only when two model classes (constant scale and location-scale generalised Gaussian) agree. It abstains on most pairs. It certifies better predictive fit, not causation.

CERT-CD (sequential adjacency certificates).

An edge is asserted only when, for every conditioning set of size at most k, a sequential test of conditional dependence has rejected. We use the nonparametric version, a sequential generalised covariance test (Shah & Peters, 2020) with spline regressions, at $k = 3$ on the chambers.

PC with a nonparametric test (PC-GCM).

PC (depth at most 3) with the same test used as PC uses a test: cross-fitted residuals, three residual products, an edge is deleted when some conditioning set yields a p-value above α. It separates the effect of the test from the effect of the sequential architecture.

BTAD (block-replicated additive discovery).

Time-ordered rows are cut into 60 contiguous blocks. A variable is added to a node's Markov blanket when its held-out gain in an additive spline model replicates across blocks (one-sided t test, Bonferroni over candidates), blankets are grown and shrunk, and edges that a small subset of the union of two blankets can explain away are removed. Its settings were fixed on simulated data ($\alpha = 0.2$, blanket at most 8).

LRG (learned replicated-gain model).

A gradient-boosting classifier on 25 features per pair (block gains with several conditioning sets, correlation, autocorrelation, within-data-set ranks), trained on 600 simulated data sets with nonlinear mechanisms, serial dependence, latent drift and duplicated sensors. The decision threshold is set on a validation split of the simulations.

PDM (pair-direction model).

A gradient-boosting classifier on 32 features (spline regression errors at three flexibilities, residual dependence, residual shape, IGCI entropies, Gaussian and location-scale log-likelihoods, the pairwise LiNGAM statistic, discreteness) trained on 40000 simulated pairs with 12 mechanism families, 6 noise laws, additive, multiplicative, location-scale and post-nonlinear structure, quantisation and serial dependence. This is the learned approach of Lopez-Paz, Muandet, Schölkopf, and Tolstikhin (2015) with different features and simulator.

4 Results

4.1 Cause–effect pairs

Figure 1 and Table 1 give the forced-choice accuracy of the learned model PDM, LOCI, a Gaussian-process ANM and the four rules on the pair sets. RECI has the highest accuracy on both chamber sets: 97.9% on the i.i.d. pairs and 94.3% on the pairs of the further recordings. The differences from IGCI (93.6%, 88.5%; exact McNemar $p = 0.29$ and 0.17) are not significant, nor is the difference from PDM on the i.i.d. pairs ($p = 0.18$); RECI is significantly more accurate than LOCI (76.6%, 68.0%), the Gaussian-process ANM, ANM-HSIC and the pairwise LiNGAM rule ($p < 0.001$ in both sets). On Tübingen LOCI has the highest unweighted accuracy (64.7%, interval $[54.6, 73.9]$, weighted 63.9%) and RECI, ANM-HSIC and the LiNGAM rule have 60.8% ($[50.6, 70.3]$; RECI weighted 70.9%, the highest weighted accuracy, with LOCI third). None of these differences is significant (exact McNemar $p \geq 0.23$ for RECI against every other method except PDM). Our LOCI run uses at most 1000 rows per pair and the authors' default settings, and its accuracy may be below published values (Section 4.1). The Gaussian-process ANM has 55.4% when its 23 exact ties (a score of zero) are counted as half correct. On the 17 Sachs pairs (which share 11 variables) every method lies between 47% and 65% and every interval contains 50% and is about 50 points wide. PDM, which reaches 95% on its held-out simulated test set and 83% on nine unseen simulated families, has 40.2% on Tübingen (interval $[30.6, 50.4]$, exact binomial $p = 0.059$ against 50%), significantly below RECI (McNemar $p = 0.002$), and is below RECI and IGCI on both chamber sets. The only pair set on which it is best is the simulated one, which is its own prior.

Recent pair rules.

None of the 2025–2026 pair rules is the best on the chamber sets, and on Tübingen and Sachs the differences are within sampling error. On Tübingen compression on the raw rows (64.7%) equals LOCI and the unweighted RECI (60.8%) is within sampling error;

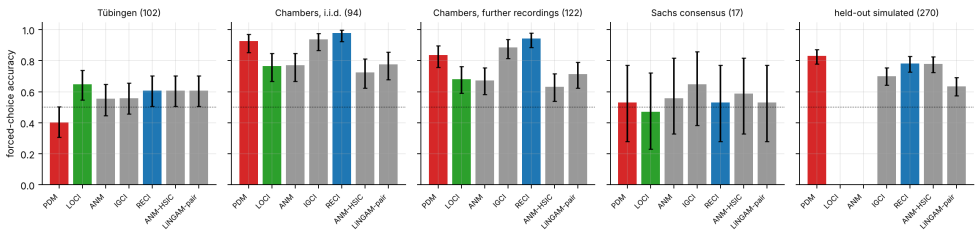


Fig. 1 Forced-choice accuracy with Clopper–Pearson intervals on five pair sets (pairs or pair-runs in parentheses). Red: PDM; green: LOCI; blue: RECI; grey: the other rules. Dotted: chance. LOCI and the Gaussian-process ANM were not run on the simulated families.

Table 1 Forced-choice accuracy (unweighted; weighted by the official weights for Tübingen). LOCI and ANM were not run on the simulated families. †Exact ties (a score of zero; 23 pairs) are counted as half correct. ‡On the 82 of 102 Tübingen pairs for which TRA returned a value. Compression and TRA are the 2025–2026 rules; compression on raw rows was run only on Tübingen and Sachs.

Method	Tübingen (102)	chambers iid (94)	chambers, further (122)	Sachs (17)	simulated (270)
PDM	0.402 (0.511)	0.926	0.836	0.529	0.830
LOCI	0.647 (0.639)	0.766	0.680	0.471	–
ANM, Gaussian process†	0.554 (0.594)	0.771	0.672	0.559	–
IGCI	0.559 (0.664)	0.936	0.885	0.647	0.700
RECI	0.608 (0.709)	0.979	0.943	0.529	0.781
ANM-HSIC	0.608 (0.609)	0.723	0.631	0.588	0.778
pairwise LiNGAM	0.608 (0.620)	0.777	0.713	0.529	0.633
Compression, raw rows	0.647 (0.659)	–	–	0.765	–
Compression, dequantised	0.608 (0.680)	0.298	0.615	0.765	–
TRA‡	0.646 (0.654)	0.862	0.615	0.588	–

dequantising the rows leaves the unweighted value at 60.8% and raises the weighted one to 68.0% (RECI: 70.9%). On the 17 Sachs pairs compression is right on 13 (76.5%, the highest of all rules, with an interval that includes chance). TRA reaches 64.6% on the pairs it scores. On the chamber pairs with independent rows RECI (97.9%) is far ahead of TRA (86.2%), and compression is at 29.8%, below chance, so the quantisation that helps on raw Tübingen rows turns the rule around on these measurements (we did not investigate why). SNOE-style orientation with the reference neighbours is right on 47.1% of the 17 Sachs arcs, 72.3% of the 47 i.i.d. chamber arcs and 76.2% of the 122 further chamber arcs, below the marginal RECI and IGCI on the same arcs.

Table 2 Skeleton precision / recall / F_1 against the reference graph (strict); mean over recordings or subsamples. Sachs: one data set, 17 reference edges. Chambers i.i.d.: two recordings, ten subsamples of 4000 rows each. Chambers with an effective sample size above 1000 (two recordings) and below 20 (seventeen recordings): the first 4000 rows in time order. Best F_1 in bold.

Method	Sachs	Chambers iid	ESS > 1000	ESS < 20
PC, $\alpha = 0.05$	1.00/0.41/0.58	0.40/0.38/0.37	0.48/0.28/0.35	0.36/0.39/0.37
GES-BIC	1.00/0.41/0.58	0.45/0.64/0.49	0.74/0.67/**0.70**	0.26/0.71/0.38
DirectLiNGAM	1.00/0.35/0.52	0.55/0.60/**0.55**	0.71/0.63/0.66	0.26/0.63/0.37
NOTEARS	1.00/0.18/0.30	0.41/0.21/0.27	0.33/0.24/0.26	0.26/0.45/0.33
DAGMA	1.00/0.29/0.45	0.44/0.33/0.36	0.44/0.41/0.39	0.25/0.52/0.34
PCMCI+	1.00/0.35/0.52	0.39/0.33/0.34	0.53/0.29/0.37	0.27/0.17/0.20
PC, nonparam. test	0.89/0.47/0.62	0.38/0.33/0.34	0.54/0.34/0.41	0.35/0.54/**0.42**
BTAD	0.89/0.47/0.62	0.56/0.36/0.41	0.82/0.46/0.58	0.60/0.16/0.25
LRG	0.90/0.53/**0.67**	0.39/0.48/0.43	0.52/0.55/0.53	0.51/0.28/0.36
BF-BIC + BOSS	0.90/0.53/**0.67**	0.43/0.63/0.49	0.57/0.61/0.58	0.30/0.68/**0.42**
CERT-CD, nonparametric	1.00/0.29/0.45	0.29/0.16/0.20	0.60/0.18/0.28	0.24/0.87/0.37

LOCI with more rows.

Raising the row cap from 1000 to 2000 (the preprocessing of the other methods) gives 65.7% on Tübingen (interval $[55.6, 74.8]$, weighted 66.4%) and 97% of the directions agree with the 1000-row run, so the cap does not explain the accuracy. We did not reproduce a higher published value and cannot rule out that the settings or the preprocessing differ from the authors'.

Abstention.

The direction certificate commits on 12 of the 80 scorable Tübingen pairs and is right on 11. Ranked by its own confidence, RECI is right on 11 of its 12 most confident pairs, and on 21 of 23 against 19 of 23 at the coverage of the constant-scale class. On the chamber pairs the certificate is right on all 31 pairs it commits to (94 pair-runs), as are the RECI and IGCI rankings at that coverage; on the simulated pairs it is right on 78 of 83 against 82 of 83 for RECI. At matched coverage the certificate is therefore not more accurate than RECI; what it adds is that it chooses its own coverage from the data. The Tübingen result of the certificate is stable over random draws of the preprocessing (9/9, 11/11, 10/10 for three further seeds), and its one error occurred on a pair with 254 rows.

4.2 Graphs

Table 2 and Figure 2 give the skeleton score on the graph benchmarks. The ranking of methods depends on the data family.

BF-BIC with BOSS.

The 2025 nonlinear score is the strongest of the recent methods on graphs. It ties LRG on Sachs (0.67: 9 of its 10 edges are in the 17-edge reference) and PC with the nonparametric test on the random-walk recordings (0.42, at 39 false edges per recording), and it is below DirectLiNGAM and GES on the i.i.d. and the larger recordings

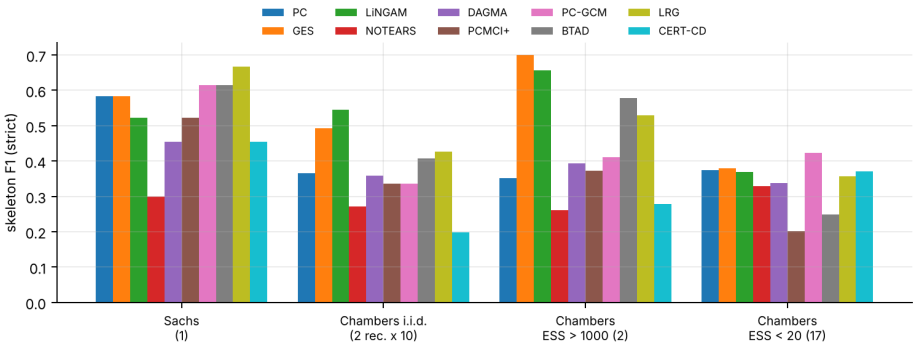


Fig. 2 Strict skeleton F_1 by data family and method.

(0.49 against 0.55 and 0.58 against 0.66–0.70). A tie on one data set of 17 edges is not evidence of superiority.

Sachs.

The methods we developed that use nonparametric components have the highest F_1 (LRG 0.67, PC with the nonparametric test 0.62–0.64, BTAD 0.62), against 0.58 for PC and GES, 0.52 for DirectLiNGAM and PCMCI+, 0.45 for DAGMA and 0.30 for the linear NOTEARS. The reference network has 17 edges, so one edge changes recall by 0.06 and the differences are not statistically distinguishable; the classical methods have the highest precision (1.00) and the new ones find a few more edges.

Chambers with independent rows.

DirectLiNGAM and GES have the highest F_1. On the two recordings with i.i.d. actuators DirectLiNGAM has $F_1 = 0.55$ and GES 0.49, against 0.32 to 0.43 for PC with either test, BTAD and LRG; on the two further recordings with a large effective sample size GES has 0.70 and DirectLiNGAM 0.66. The actuators there are uniform and the mechanisms may be nearly additive, which would suit DirectLiNGAM (we did not test this explanation); each group has two recordings and no interval can be given. BTAD has the highest precision (0.56, against 0.55 for DirectLiNGAM, and 0.82) at lower recall. The sequential certificate has the lowest F_1 on the i.i.d. recordings (0.20).

Random-walk recordings.

On the seventeen recordings with an effective sample size below 20 every method has a strict F_1 between 0.20 and 0.42; the best, PC with the nonparametric test, has precision 0.35. BTAD has the highest precision (0.60, with 2.5 false edges per recording against 16 to 52 for PC, GES and DirectLiNGAM) and recall 0.16. PCMCI+, the method built for autocorrelated series, has recall 0.17 and $F_1 = 0.20$ there, as expected when the recordings contain no usable lag structure (Section 4.3); DAGMA has $F_1 = 0.34$. Paired Wilcoxon tests over the 15 wind-tunnel recordings, with Holm correction over the eleven comparisons, show that BTAD's F_1 is lower than that of

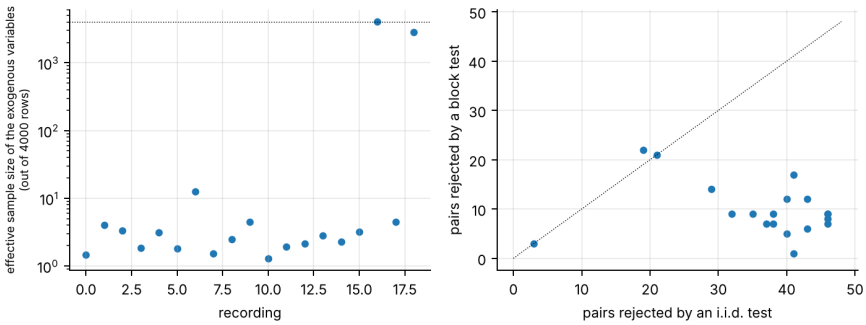


Fig. 3 Left: effective sample size (log scale) of the exogenous variables in each of the nineteen new recordings. Right: number of non-adjacent pairs whose graph-implied independence is rejected by an i.i.d. test and by a block test.

ten of the other methods (adjusted $p \leq 0.02$) and higher than that of PCMCI+; LRG is not distinguishable from PC, GES or DirectLiNGAM.

4.3 The walk recordings carry a few independent observations

Figure 3 shows why the walk recordings defeat every method. In the fifteen wind-tunnel walks the mean lag-1 autocorrelation of the exogenous variables is 0.998 and the implied effective sample size is about 3 of 4000 rows (range 1.3 to 12.5). The mean cross-correlation between a load and the rotation speed it drives varies by only 0.012 for lags from -6 to $+6$ (0.974 to 0.985), so the recordings also contain no usable lag structure. An i.i.d. test of the independences implied by the graph rejects 40.8 non-adjacent pairs per recording on the wind-tunnel walks and a test on 40 contiguous blocks rejects 8.5, a ratio of 4.8; over the seventeen recordings with an effective sample size below 20 the ratio is 4.1 (38.9 against 9.5), over all nineteen 3.7, for the pressure-control recording 2 and for the light-tunnel walk 1 (21 against 21). The block test has lower power by construction, so a smaller number of rejections does not by itself show that the i.i.d. rejections are artefacts; it does show that a reference set built with the i.i.d. test has many more pairs than one built with a test that respects serial dependence, and that scores adjusted for it should be read with caution. Two recordings of the new set (`actuators_white`, `smooth_polarizers`) have an effective sample size above 1000, and on these two the classical methods have the highest F_1; two recordings are too few to separate effective sample size from device. We did not run a controlled test (thinning, or a time-series-aware test on the same recordings), which would be the direct evidence.

4.4 Where the new methods help: monitoring and nonlinear mechanisms

On simulated graphs the sequential certificate does what it was designed for (Table 3). When each method is rerun at every sample size from 100 to 6400 on 150 random eight-node graphs and a graph counts as a failure if any estimate contains a false edge, CERT-CD fails in 2.7% of the graphs under Laplace errors, against 27% to 91% for

Table 3 Probability that a false edge is ever reported over $n = 100, \ldots, 6400$ (150 graphs, $d = 8$), and graphs with a false edge under nonlinear mechanisms (60 graphs).

Method	Laplace errors	Gaussian errors
CERT-CD, linear test	0.027	0.040
CERT-CD, nonparametric test	0.060	0.047
PC, $\alpha = 0.05$ / 0.01	0.747 / 0.267	0.773 / 0.320
PC, nonparametric test, $\alpha = 0.05$ / 0.01	0.767 / 0.387	0.787 / 0.320
GES-BIC	0.913	0.960
DirectLiNGAM	0.760	1.000
nonlinear mechanisms: linear test (of 60)	33	28
nonlinear mechanisms: nonparametric test (of 60)	1	2

Table 4 Learned models: simulated test against real data.

Model	simulated test	real data	measure
PDM	0.95 / 0.83	0.40 Tübingen; 0.93, 0.84 chambers	forced accuracy
LRG	0.86	0.67 Sachs; 0.43, 0.53, 0.36 chambers	skeleton F_1

PC, GES and DirectLiNGAM and 39% to 77% for PC with the same nonparametric test, which is consistent with the delete-on-non-rejection architecture, and not the test, being responsible (a single comparison). With nonlinear mechanisms the linear test of the certificate fails in 33 of 60 graphs under Laplace errors and 28 of 60 under Gaussian errors, and the nonparametric version in 1 and 2 of 60. The guarantee of the linear version is exact only for Gaussian regression errors, and that of the nonparametric version is asymptotic. The table shows false edges only, not power: the recall of the nonparametric certificate is 0.00 at $n = 100$ and 0.05 at $n = 200$ under Laplace errors, and DirectLiNGAM is run on Gaussian errors, which violate its assumption. These are simulations within the models; on the chambers the false-edge control did not hold (a false edge adjusted for a reference set built with an i.i.d. test in 12 of 19 recordings, a strict false edge in 18, with shuffled rows).

4.5 The gap between simulation and measurement

The two learned models are accurate on data from their training distribution and lose that on real data (Table 4). PDM falls from 0.95 to 0.40 on Tübingen; LRG falls from an F_1 of 0.86 on simulated graphs to 0.36–0.43 on the chambers, and its within-recording ranking of pairs on the wind-tunnel walks is close to chance (mean AUC 0.55 over 15 recordings, range 0.51 to 0.60). We cannot say which feature of the real data is responsible; the simulators contain serial dependence, latent drift, duplicated sensors and several mechanism families and still do not cover them.

5 Discussion

What the evidence supports.

On the real data we examined, simple methods are hard to beat. On the chamber pairs RECI, which compares two regression errors, has the highest accuracy, not distinguishable from IGCI; on Tübingen no method is distinguishable from RECI except PDM, which is worse. On graphs the highest F_1 on recordings with independent rows is DirectLiNGAM and GES. Of the four best-F_1 cells, two belong to methods we developed (LRG on Sachs, 0.67, and PC with the nonparametric test on the walk recordings, 0.42) and in both the 2025 BF-BIC score with BOSS ties them; the other 2025–2026 rules (compression, TRA, SNOE-style orientation) do not lead any chamber set, by margins of one or two edges and of 0.04 that are within sampling error. The methods we developed show their intended benefit in simulation and some of it on measurements: lower false-edge rates under repeated inspection and nonlinear mechanisms in simulation, and the highest precision on the chamber recordings, BTAD, at the price of recall. The two classifiers trained on simulated data are the clearest negative results: a model that is nearly perfect on a broad simulator can be below chance on real pairs.

Implications for evaluation.

First, the effective sample size of a recording should be reported: seventeen of nineteen new chamber recordings had fewer than twenty effective observations, which no test that assumes independent rows can cope with, and which also contaminates reference sets built with such tests. Second, a score adjusted for dependences the reference graph cannot explain, if used, should be shown next to the strict score, with the construction of the adjustment disclosed and independent of the methods; ours was not independent of the test families used by some methods. Third, the protocol matters: some of our early results came from choices made after seeing the data, and we report which numbers are frozen and which are not (Section 2.5).

Limitations.

The reference networks are small, the Sachs consensus is contested, and the chamber graphs are bipartite, so recovery of the skeleton is mostly feature selection from actuators to measurements. Most differences between graph methods are within sampling error, and the effective sample size of the walk recordings limits what any method can show. We did not evaluate amortised methods (AVICI), CAM or QCCD, and several baselines are re-implementations or default runs (LOCI on at most 1000 rows); a better-tuned baseline could change a ranking. The Causal Chambers also contain interventional experiments, where a two-sample shift test recovers the descendants of an intervened variable with no false positives on 58 experiments (88 found, 20 missed; `scripts/audit_followups.py`), and we did not use them for the evaluation. The pair sets are structurally easy (actuator causes with simple marginals) and the chamber pair-runs are not independent. We did not evaluate RCC or NCC, Slope, CAM, nonlinear continuous-optimisation methods, GIES or ICP, and we applied no multiplicity control except in the BTAD comparison. The new methods were developed by the

author of this paper, which is a source of bias; the frozen protocol and the public code are the mitigation.

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Data and code availability. All simulation code, experiment scripts, random seeds and figure-generating scripts are available at https://github.com/vinhqdang/SPRINT-CD. The protein-signalling data of Sachs et al. (2005) are redistributed with the `bnlearn` package (Scutari, 2010) and are downloaded by the experiment script rather than included in the repository. The Tübingen cause-effect pairs of Section 2 are downloaded on first use from https://webdav.tuebingen.mpg.de/cause-effect/ and are not redistributed.

Author contributions. The author is solely responsible for all aspects of this work.

Use of generative AI. A large language model was used in support of writing this manuscript. The author verified and modified the resulting content and takes full responsibility for the manuscript in its entirety, including its claims, derivations, experiments and references.

Appendix A Method details

Pair features.

The direction hypothesis $c \to e$ is described by the log cross-validated mean squared error of additive cubic spline regressions of e on c with 2, 4 and 8 knots, the log HSIC between the residual and the cause, the share of $\log r^2$ explained by c, the skewness and kurtosis of the residual, the correlation of r^2 with c^2, held-out Gaussian and location-scale log-likelihoods, and the spacing entropy, skewness and kurtosis of c; the feature vector is the difference between the two directions plus their sum and symmetric nuisance features (sample size, correlation, share of distinct values). The classifier has depth 6, 500 trees and learning rate 0.05.

Graph features (LRG).

For every pair, the held-out gain of adding one variable to a model of the other with 40 contiguous blocks (mean, t statistic, share of positive blocks), in both directions, with an empty conditioning set, the three strongest other predictors, and the single conditioner that most explains the pair away; the absolute correlation; the smaller and larger lag-1 autocorrelation of the two variables; the log number of variables; and within-data-set ranks of the three t statistics. Depth 4, 300 trees. The decision threshold (0.4) maximises validation F_1.

BTAD.

Spline ridge models with three knots at the quartiles, inputs clipped at the 1% and 99% quantiles, pairwise products when the conditioning set has two or more members; gap of 5% of the block length between blocks; 60 blocks; Bonferroni over the candidates of each step; blanket of at most 8 variables; spouse removal over subsets of size at most 2 of the union of the blankets.

PC with a nonparametric test.

Residuals of X_v and X_v^2 on an additive spline of X_S with two-fold cross-fitting; the three products $r_i r_j$, $r_i r_j^{(2)}$ and $r_j r_i^{(2)}$; p-value $= \min(1, 3 \min_k p_k)$.

Sequential certificates.

The adjacency certificate and the direction certificate are described, with their assumptions, in the companion paper (`paper/certcd2` in the repository); here only their outputs are used.

Reproduction.

Each experiment is a script in `experiments/`, committed before it was run, with a summary script in `scripts/`; the figures are made by `scripts/make_benchmark_figs.py`. All results are in `results/` as JSON. Seeds are fixed. The Chamber data and the Tübingen pairs are downloaded on first use.

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